

Exploratory Data Analysis of U.S. Airline Operational Delays

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This Exploratory Data Analysis (EDA) project investigates operational delays within U.S. airline operations using a large dataset containing airline performance information for January 2024. The primary objective of this analysis is to identify operational patterns, understand the major causes of delays, and evaluate airline performance across different carriers.

The airline industry operates in a highly complex and interconnected environment where delays can propagate rapidly throughout the network. Operational disruptions may originate from weather events, air traffic congestion, aircraft rotation delays, maintenance issues, or carrier-related operational inefficiencies. Understanding these factors is critical for improving operational performance and minimizing passenger disruption.

- This project explores several key operational questions:
- What is the overall distribution of airline arrival delays?
- Which airline carriers experience the highest average delays?
- What operational factors contribute most significantly to delays?
- How strongly are departure delays related to arrival delays?
- What operational patterns can be identified through correlation analysis?

The analysis was conducted using Python, Pandas, Matplotlib, and Seaborn within a Jupyter Notebook environment.

Dataset Overview

The primary dataset used for this analysis contains operational flight data for January 2024. To maintain manageable processing times and improve exploratory analysis efficiency, the project focused on a single month of operations while still retaining a sufficiently large dataset for meaningful analysis.

Dataset Characteristics

Characteristic	Value
Total Records	547,271 flights
Total Columns	11 variables
Time Period Covered	January 2024
Data Type	Airline operational performance data

Key Variables Used

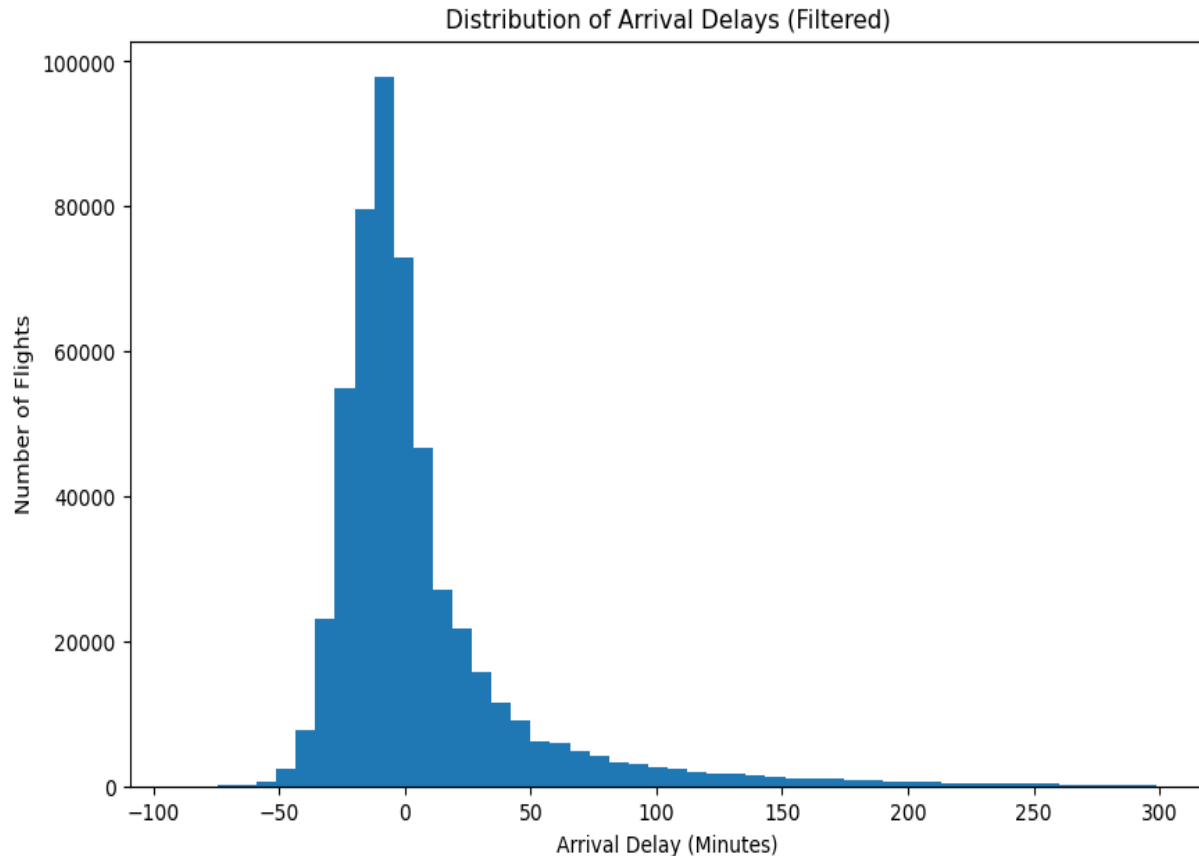
Variable	Description
FL_DATE	Flight date
OP_UNIQUE_CARRIER	Airline carrier code
ORIGIN	Origin airport
DEP_DELAY	Departure delay in minutes
ARR_DELAY	Arrival delay in minutes
CANCELLED	Flight cancellation indicator
CARRIER_DELAY	Carrier-related operational delays
WEATHER_DELAY	Weather-related delays
NAS_DELAY	National Airspace System delays
SECURITY_DELAY	Security-related delays
LATE_AIRCRAFT_DELAY	Delays caused by late inbound aircraft

Data Quality Observations

Several delay-related columns contained missing values. However, these missing values appear to reflect operational logic rather than poor data quality. Delay categories such as weather delay, NAS delay, and security delay are likely populated only when those specific operational disruptions occur.

Additionally, the dataset contained several extreme outlier values in delay variables. These outliers were filtered in certain visualizations to better observe the primary distribution patterns.

1. Distribution of Arrival Delays



Analysis

The histogram of arrival delays demonstrates that most flights arrived relatively close to their scheduled arrival times. A large concentration of flights clustered near zero minutes indicates that the majority of operations remained relatively punctual.

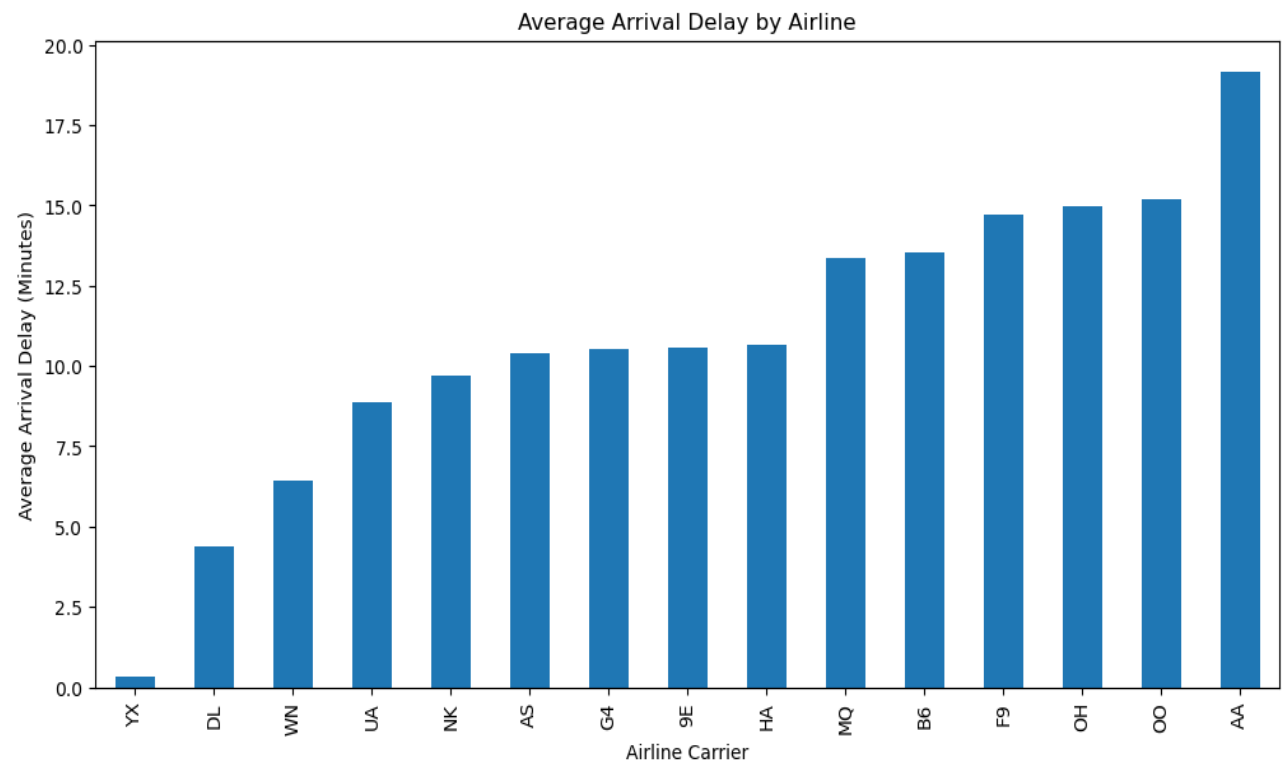
The dataset also contains many negative arrival delay values, indicating flights arriving earlier than scheduled. This may result from conservative block time scheduling, favorable weather conditions, or efficient operational performance.

The distribution is heavily right-skewed, with a long positive tail representing severe operational delays. These outliers suggest the presence of irregular operational events capable of producing substantial downstream disruptions.

Key Insight

Although most airline operations remained relatively stable, a smaller number of severe operational disruptions significantly impacted overall system performance.

2. Average Arrival Delay by Airline Carrier



Analysis

The analysis revealed noticeable differences in average arrival delay performance among airline carriers. Certain carriers maintained relatively low average delays, while others experienced substantially higher operational delays.

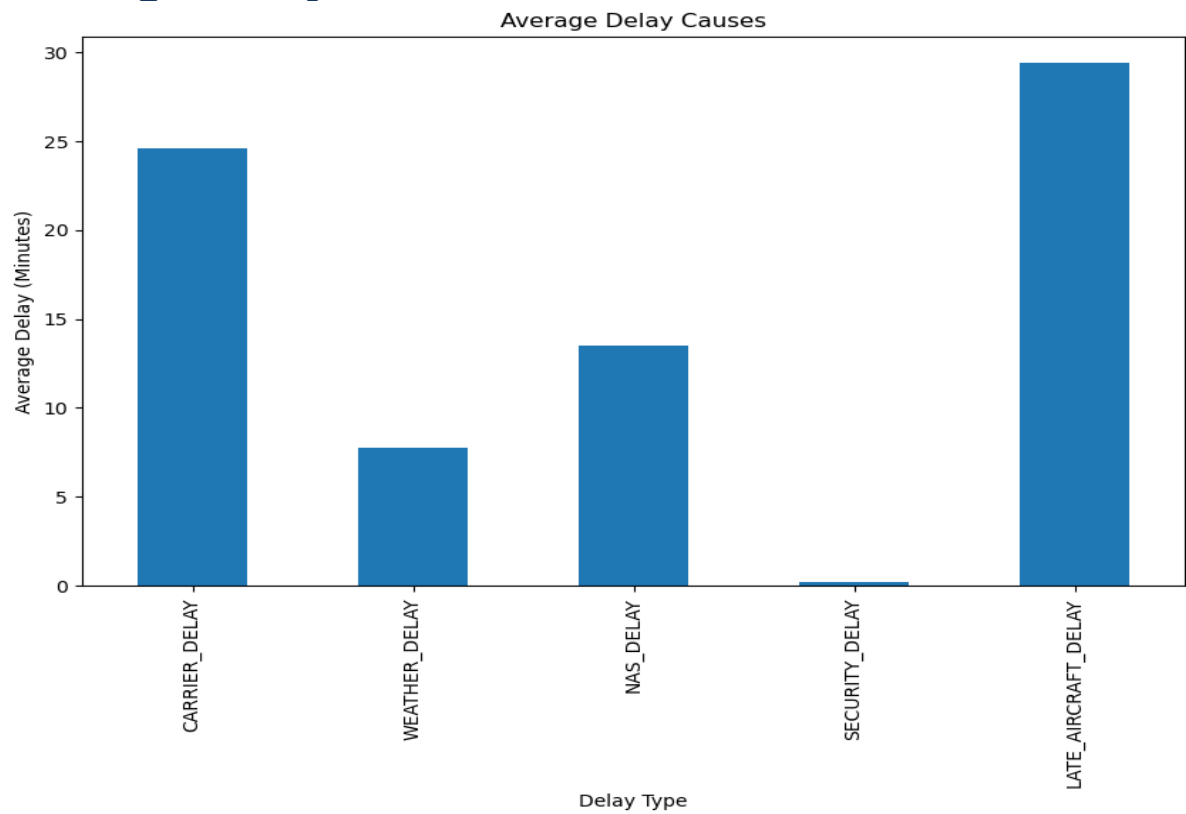
Airlines such as YX and DL demonstrated lower average delay performance, while carriers such as AA exhibited significantly higher average arrival delays.

These variations may reflect differences in route network complexity, hub congestion exposure, aircraft utilization strategies, operational recovery efficiency, and flight scheduling practices.

Key Insight

Operational performance varies substantially across airline carriers, suggesting that network structure and operational management practices play an important role in delay performance.

3. Average Delay Causes



Analysis

The delay cause analysis revealed that Late Aircraft Delay was the largest contributor to operational delays, followed closely by Carrier Delay.

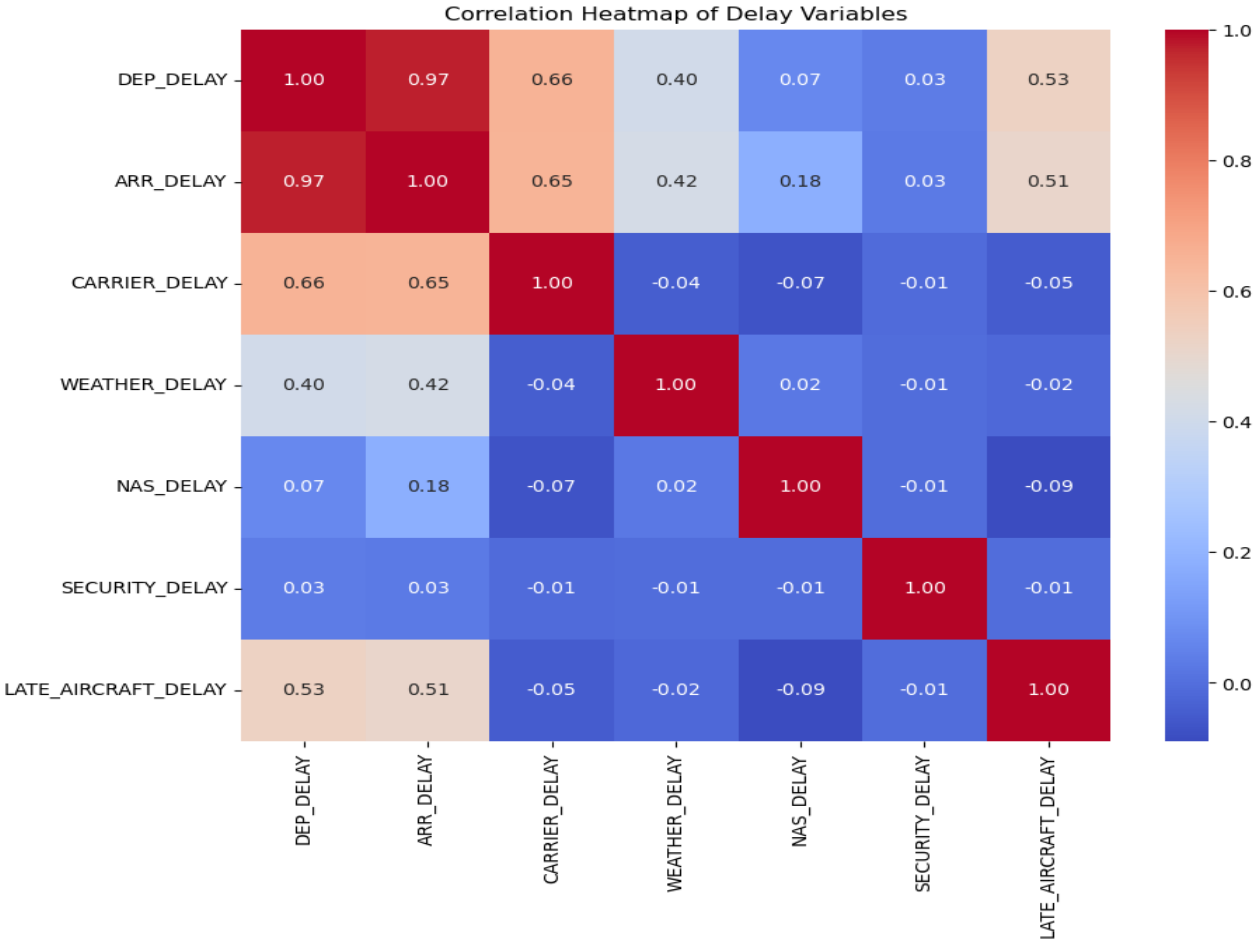
Late aircraft delays indicate operational delay propagation, where delays from previous flights impact subsequent aircraft rotations throughout the airline network.

Carrier-related delays may include maintenance issues, crew scheduling problems, turnaround inefficiencies, gate availability issues, and operational coordination delays.

Key Insight

Operational delay propagation and internal airline operational inefficiencies appear to contribute more significantly to delays than weather-related disruptions.

4. Correlation Heatmap of Delay Variables



Analysis

The correlation heatmap revealed an extremely strong relationship between departure delays and arrival delays, with a correlation coefficient of 0.97.

This finding indicates that delays occurring during departure operations strongly propagate throughout the flight lifecycle and directly impact arrival performance.

This suggests that operational recovery efforts should prioritize departure-phase disruptions before delay propagation spreads through downstream flight operations.

Key Insight

Operational disruptions occurring during the departure process tend to propagate throughout the entire operational network, emphasizing the importance of early operational recovery and efficient turnaround management.

Business Impact

Operational Relevance

- Improved turnaround management may reduce propagated delays throughout the operational network.
- Airlines could prioritize gate efficiency and aircraft rotation planning to improve schedule reliability.
- Predictive delay monitoring systems may help identify disruptions earlier and support proactive operational recovery.
- Strong departure-to-arrival delay propagation suggests that early operational intervention is critical for minimizing downstream disruption.

Key Findings

1. Most flights operated relatively close to schedule, although severe operational outliers were present.
2. Arrival delay distributions were heavily right-skewed, indicating that a smaller number of severe disruptions significantly impacted operational performance.
3. Significant performance variation existed between airline carriers.
4. Late aircraft delays and carrier-related delays were the largest contributors to operational disruptions.
5. Departure delays demonstrated an extremely strong relationship with arrival delays, suggesting strong delay propagation effects.
6. Weather disruptions contributed to delays but were not the dominant operational factor in this dataset.

Challenges and Limitations

- The dataset covers only one month of operations, limiting the ability to analyze seasonal trends.
- Extreme outlier delay values distorted certain visualizations and required filtering for clearer interpretation.
- Several delay-related columns contained significant missing values. However, these missing values appear operationally valid because delay categories are populated only when applicable.
- Airline carrier codes were abbreviated, requiring operational interpretation rather than direct airline naming.

Conclusion

This Exploratory Data Analysis successfully identified meaningful operational patterns within U.S. airline operations.

The findings suggest that operational delay propagation, aircraft rotation efficiency, and carrier-related operational factors play a major role in airline performance. The strong relationship between departure delays and arrival delays highlights the importance of efficient operational recovery and proactive disruption management.

The project demonstrates how data analytics can be used to investigate operational performance, identify systemic inefficiencies, and support data-driven decision-making within the airline industry.

Tools Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook
- Anaconda

References

- U.S. Airline Operational Performance Dataset
- Python Documentation
- Pandas Documentation
- Matplotlib Documentation
- Seaborn Documentation